

ECE 649 — Generative AI

Fall 2026 • 3 Credits

Wednesdays 11:15 AM–2:00 PM • MEB 409

Instructor	Dr. Willie McClinton
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Class Format	One weekly 165-minute meeting: Block A (70 min), 10-minute break, Block B (70 min), 15-minute wrap-up / Q&A.
Semester Dates	Classes begin Aug. 17; this course first meets Aug. 19. Classes end Dec. 1; Reading Day Dec. 2; final exams Dec. 3–9, 2026.

Course Description

This course introduces generative AI as a new generation of AI systems that create new objects, including text, code, images, audio, video, robot behaviors, 3D shapes, and molecules. Students learn the foundations of AI, machine learning, and deep learning before studying modern generative methods such as transformers, large language models, diffusion models, flow matching, retrieval-augmented generation, and agentic workflows. The course emphasizes both theory and implementation through an interactive course repository, in-person quizzes, one mid-term examination, one cumulative final examination, and a final project in which students build and present a generative AI application.

Course Objectives

- Explain the core ideas, vocabulary, and mathematical foundations behind modern generative AI systems.
- Understand and apply major generative model families, including transformers, LLMs, diffusion, and flow matching.
- Build generative AI for text, code, images, audio, video, robotics, 3D shapes, molecules, agents, and RAG systems.
- Evaluate generative AI systems for quality, reliability, safety, bias, hallucination, and real-world deployment constraints.
- Use industry-standard AI coding assistants together with GitHub, Markdown, and project workflows to build, document, and extend course projects.

Course Organization: Six Parts

Part I	Foundations of AI and Generative Systems
Part II	Text and Code Generation with Transformers and Large Language Models
Part III	Non-Text Generation with Diffusion and Flow Matching: Image, Audio, and Video
Part IV	Advanced Topics I — Robotics and Simulators
Part V	Advanced Topics II — 3D Shapes and Molecules
Part VI	Advanced Topics III — Agents, RAG, and OpenClaw

Each Part occupies two consecutive teaching weeks and therefore receives four ~70-minute lecture blocks. Weeks 13–14 are reserved for synthesis, review, project refinement, and final presentations.

Assessment

Component	Weight	Schedule
In-person quizzes & participation	20%	Throughout semester
Mid-term examination	20%	Fri., Oct. 9, 2026 — outside regular lecture blocks; delivery/logistics announced in class
Cumulative final examination	20%	University final-exam period: Dec. 3–9, 2026; exact Registrar-assigned date/time TBA

Final generative AI project and presentation — 40% total:

- Project proposal due: Wednesday, Oct. 7, 2026 at 11:59 PM.
- Final project presentation: Wednesday, Nov. 18, 2026 during the regular class meeting.
- The project must include a clear problem statement, architecture, data or knowledge source, evaluation method, and discussion of limitations.

Weekly Meeting Structure

The regular Wednesday meeting runs 11:15 AM–2:00 PM (165 minutes): 11:15–12:25 Block A; 12:25–12:35 break; 12:35–1:45 Block B; 1:45–2:00 wrap-up, questions, quiz/project logistics, and preview.

Fall 2026 Lecture Calendar

The dates below use the University of Miami Fall 2026 Academic Calendar. Fall Recess is Oct. 10–13; Thanksgiving Recess is Nov. 24–29; Dec. 2 is Reading Day. Accordingly, ECE 649 has 14 regular Wednesday class meetings.

Wk	Date	Part	Block A (11:15–12:25)	Block B (12:35–1:45)
1	Aug. 19	I — Foundations	What is Generative AI? Historical context, generative vs. discriminative approaches, major modalities and model families.	Probabilistic ML basics: random variables, distributions, conditional probability, likelihood, MLE, entropy, cross-entropy, KL divergence.
2	Aug. 26	I — Foundations	Formulating the generative AI problem: learning distributions, sampling, conditional generation, density estimation, controllability and evaluation.	Latent representations and representation learning: embeddings, latent spaces, autoencoders, VAEs, representation quality.
3	Sept. 2	II — Text & Code	Historical context and first attempts in text modeling: n-grams, statistical language models, RNNs/LSTMs, sequence-to-sequence models and limitations.	Recasting text as a generative AI problem: tokenization, autoregressive factorization, next-token prediction, training vs. generation.
4	Sept. 9	II — Text & Code	Transformers and LLMs: attention, Q/K/V, self-attention, transformer blocks, positional information and decoder-only language models.	Breakthroughs with scale: pretraining, scaling, zero/few-shot behavior, instruction tuning, preference optimization, prompting and code generation.
5	Sept. 16	III — Diffusion & Flow	Image generation and first attempts: feature inversion/style transfer, autoencoders/VAEs, GANs and autoregressive image models.	Diffusion setup: iterative refinement, forward noising, denoising, UNets, conditioning, training and sampling intuition.
6	Sept. 23	III — Diffusion & Flow	Diffusion formalized: forward/reverse processes, noise schedules, denoising objectives, score interpretation and sampling.	Flow matching: normalizing-flow intuition, continuous dynamics, probability paths, vector fields and modern flow-matching models.
7	Sept. 30	IV — Robotics & Simulators	Generative AI for robotics: action/trajectory generation, imitation learning, generative policies and diffusion policies.	Vision-language-action and embodied foundation models: language-conditioned behavior, multimodal representations and robot control.
8	Oct. 7	IV — Robotics & Simulators	World models and simulators: learned dynamics, generative environments, synthetic experience and model-based reasoning.	Generative robotics in practice: simulation, data generation, sim-to-real, safety and evaluation. Project proposal due 11:59 PM.
9	Oct. 14	V — 3D & Molecules	3D representations and generation: meshes, point clouds, voxels, neural fields, NeRFs and Gaussian splats.	Text-to-3D and image-to-3D: reconstruction, generative design, scene/world generation and evaluation.
10	Oct. 21	V — 3D & Molecules	Molecular generation: graphs, SMILES, molecular representations, validity, novelty and property conditioning.	Generative science: proteins, structure prediction, diffusion/flow methods for molecules, drug discovery and scientific workflows.
11	Oct. 28	VI — Agents, RAG & OpenClaw	Retrieval-augmented generation: ingestion, chunking, embeddings, vector search, reranking, context construction and citations.	Building and evaluating RAG systems: retrieval quality, grounded generation, failure modes, evaluation and practical architectures.
12	Nov. 4	VI — Agents, RAG & OpenClaw	AI agents: tools, planning, reasoning, memory, execution, critics, feedback loops and human-in-the-loop supervision.	Agentic systems in practice: coding/research agents, multi-agent workflows, computer use, OpenClaw-style workflows and reproducibility.
13	Nov. 11	Integration & Recap	Synthesis I: Parts I–III — probability, latent variables, autoregression, transformers, diffusion and flow matching as alternative generative formulations.	Evaluation, reliability, safety and trustworthy GenAI; project refinement and final-exam preparation.
14	Nov. 18	Integration & Recap	Synthesis II: Parts IV–VI — embodied generation, 3D/science, retrieval and agents; connecting the full course.	Final project presentations; cumulative course takeaways and final-exam review.

Calendar notes: No Wednesday class is scheduled during Thanksgiving Recess (Nov. 25), and Dec. 2 is the University Reading Day. The Fall 2026 calendar states that final examinations occur Dec. 3–9. The exact ECE 649 final-exam date/time should be updated once the University publishes/assigns the official final-exam schedule.

Textbooks and Resources

- Omar Sanseviero, Pedro Cuenca, Apolinário Passos, and Jonathan Whitaker, Hands-On Generative AI with Transformers and Diffusion Models, O’Reilly Media.
- Kevin P. Murphy, Probabilistic Machine Learning: An Introduction, MIT Press.
- Kevin P. Murphy, Probabilistic Machine Learning: Advanced Topics, MIT Press.
- Course GitHub repository containing lecture notes, assignments, examples, MDX lessons, project templates, and AI-agent-readable study materials.
- Selected research papers, technical blogs, model cards, documentation, and open-source repositories assigned throughout the semester.

Indicative Reading Alignment for Parts I–III

Part	Primary alignment	Use
I — Foundations	Hands-On Chs. 1 & 3; Murphy Introduction Chs. 1–2, 4, 6, 20; Murphy Advanced Chs. 20–21	GenAI framing, probability, latent representations, autoencoders/VAEs.
II — Text & Code	Hands-On Ch. 2; Murphy Introduction Ch. 15; Murphy Advanced Ch. 22	Sequence modeling, autoregressive generation, attention, transformers and LLMs.
III — Diffusion & Flow	Hands-On Ch. 4; Murphy Advanced Chs. 23–25; supplemental flow-matching readings	Image-generation history, diffusion mathematics, scores/flows and flow matching.

Course Assessment and Integration

The final portion of the course emphasizes integration, testing, project refinement, final presentations, and preparation for the cumulative final examination. The final examination is cumulative across Parts I–VI.

Schedule Authority and Changes

The University academic-calendar dates incorporated here are based on the Fall 2026 Academic Calendar (updated May 27, 2026), which is marked “Subject to Change.” Course topics, readings, and assessment logistics may be adjusted by the instructor as needed; any changes will be communicated to students through the official course channels.